Large Language Models for Logical Fallacy Detection

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Abstract. Identifying logical fallacies is essential for maintaining logical reasoning and reducing false information in a variety of domains, such as the media, law, and education. We present an extensive study on the use of large language models (LLMs) for logical fallacy detection and provide a comparative overview of model performance across various fallacy classes. We evaluate the logical fallacy detection capabilities of multiple state-of-the-art models (LLaMA, Qwen, Gemma, Phi) utilizing accuracy, precision, recall, and F1-score as assessment measures. According to our findings, our models do well on simple fallacies like "circular reasoning," but they have trouble with more interpretive reasoning when it comes to more complex categories like "equivocation" and "intentional". These results highlight the potential of LLMs in fallacy detection tasks but also indicate a need for improved prompt engineering, fine-tuning, and context-rich datasets to enhance interpretive accuracy. This research offers insights into advancing LLMs for critical reasoning applications, contributing to improved information integrity across domains.

Keywords: Large language models · Logical fallacies · Reasoning.

1 Background

The rapid advancement of large language models (LLMs) has opened new avenues for natural language processing, allowing for exceptional performance on a variety of downstream tasks [12, 6]. In addition, it has also expanded applications spanning from conversational agents to complex reasoning tasks [2, 17, 3]. Among these tasks, logical fallacy detection has emerged as a critical area of focus, given its implications for the integrity of information and reasoning processes in various domains, including education, law, and media literacy. Logical fallacies undermine sound argumentation and can lead to flawed conclusions, making the ability to identify and address these fallacies essential.

A fallacy is defined as "a mistake in reasoning [...] that occurs with some frequency in real arguments and which is characteristically deceptive" [5]. The task of classifying fallacies, or developing a classification model to distinguish between various forms of fallacies, has been taken up by recent publications [1]. [13] combined methods like large transformer language models and the Structure Aware Hypothesis [29] from natural language inference. In this line of study, the underlying logical shape of the text is extracted via argument mining [15], and the resulting features are used to train a classifier. Interestingly, the datasets presented by [13] only include false assertions, which means that they may only be classified as fallacious types if the argument is presumed to be false. More recently, [16] focused on assessing the effectiveness of GPT-4, in identifying logical fallacies as part of a digital misinformation intervention. This evaluation is particularly important because logical fallacies are a common feature in misinformation, and understanding the capabilities of LLMs in this domain could aid in combating such content. They found that while GPT-4 showed strong performance in identifying logical fallacies, its performance varied across different types of fallacies.

This study investigates the ability of leading LLMs—LLaMA, Qwen, Gemma, and Phi—to detect logical fallacies using metrics like accuracy, F1-score, recall, and precision. We also analyze their sensitivity to different fallacy types to better understand their reasoning processes. To enhance interpretability, we apply methods from interpretable AI, offering insights into model decisions and increasing transparency and accountability. By providing justifications for classifications, we aim to clarify how LLMs identify fallacies and support their use in fostering critical thinking and informed decision-making.

Our work aims to present a thorough analysis across different LLMs for the identification of logical fallacies, together with new methods for evaluating the interpretability and effectiveness of these models. The primary contributions we have made in this paper are:

1. The paper presents a comprehensive analysis of state-of-the-art LLMs (LLaMA, Qwen, Gemma, Phi) for detecting various types of logical fallacies, providing insights into their strengths and weaknesses.
2. The study highlights how LLMs perform differently across fallacy categories, showing strong results on simpler fallacies like "circular reasoning" while facing challenges with more complex types such as "equivocation" and "intentional."
3. The research suggests enhancements in prompt engineering, fine-tuning, and dataset design, offering practical guidance for improving LLMs' ability to detect and interpret complex logical fallacies.

2 Related Work

Logical Reasoning. Recent developments have demonstrated that LLMs may generate intermediate reasoning stages to carry out reasoning tasks efficiently [23, 26, 28]. Formal reasoning, which is frequently applied in logic and mathematics,

is a methodical, logical process that adheres to a set of norms and principles. Informal reasoning, which is more prevalent in daily life, is a less formal method of problem solving that depends on experience, common sense, and intuition to reach conclusions. Informal reasoning is more flexible and open-ended but may also be less reliable than formal reasoning, which is more ordered and dependable [10]. The logical reasoning skills of large language models are gaining more and more attention from researchers [9, 27, 31, 21]. For example, [19] integrated first-order logic with large language models. [20] and [30] equipped extensive language models with symbolic solutions. An adversarial pre-training approach was presented by [22] to improve logical thinking. Multi-step explicit planning was added to the inference process by [30]. A contrastive learning strategy was put up by [11] to enhance logical question responding.

Logical Fallacies. False reasoning patterns are known as logical fallacies [4]. They are frequently recognized since their claims are not backed up by any proof. Finding logical fallacies in natural language text has a wide range of possible uses, including tracking false information and confirming assertions. Furthermore, being able to understand erroneous arguments can improve the logic and educational value of speech. Fallacies can also be deliberately committed in order to mislead or influence others, and have been used to disseminate propaganda in news articles [18]. Recent attempts have been made to categorize fallacies using deep learning models. For example [13] proposed a dataset, where the fallacies were categorized using a structure-based approach. Follow up work has also tried to classify logical fallacies and valid statements [14], and build a software that addresses question-answering fallacies [7]. [24] investigated discussion answers for the ad hominem fallacy. In their study, [8] investigated the ad hominem fallacy in online arguments. Argumentation essays were found to have weak arguments by [25]. A multi-task prompting technique was presented by [1]. It allows users to collaboratively learn fallacies from several datasets. To our knowledge, no prior work has tried to test out the logical fallacy detection capabilities of recent open source LLMs.

3 Methodology

In this study, we evaluate the logical fallacy detection capabilities of different state-of-the-art large language models, such as LLaMA, Qwen, Gemma, and Phi. These models were selected based on their recent advancements in natural language understanding and generation, as well as their potential to address complex reasoning tasks such as logical fallacy detection. Each model was evaluated in the context of its ability to identify fallacies across a range of types, including simple and complex categories.

This section also outlines the data used for evaluation, data pre-processing, evaluation and implementation, the baselines for comparison, and the specifics of the implementation.

3.1 Dataset

We make use of the LOGIC and LOGIC Climate logical fallacy datasets from [13], which has 2,449 instances of logical fallacies spanning 13 different categories. The LOGIC dataset comprises of typical instances of logical fallacies gathered from several online educational resources designed to instruct or assess students' comprehension of logical fallacies. The logical fallacy examples was gathered manually from Google, totaling about 600 samples, and automatically gathered from three student quiz websites: Quizziz, study.com, and ProProfs [13]. The LOGIC Climate dataset contains more difficult examples of the same logical error types on the subject of climate change. It comprises of 1,079 samples of logical fallacies, with an average of 35.98 tokens per sample and a vocabulary of 5.8K words. A sample of the dataset structure is illustrated in Table 1. The full dataset contains 13 logical fallacy types, each represented by exemplar statements that demonstrate the reasoning error characteristic of that fallacy category.

Table 1: Structure of the LOGIC Dataset

Text	**Label**
"You're too ugly to be class president!"	Ad Hominem
Everyone is going to get the new smart phone when it comes out this weekend. Why aren't you?	Ad Populum
Do the best for your baby	Appeal to Emotion
America is the best place to live, because it's better than any other country.	Circular Reasoning
Noisy children are a real headache. Two aspirin will make a headache go away. Therefore, two aspirin will make noisy children go away.	Equivocation
I am committed to preserving traditional marriage, the union of one man and one woman.	Fallacy of Credibility
The Senator thinks we can solve all our ecological problems by driving a Prius.	Fallacy of Extension
Most Arabs are Muslims and all the 9/11 hijackers were also Muslims. Therefore most Arabs are hijackers.	Fallacy of Logic
"I believe deeply that canceling the fashion show is the best decision. I know it is. I am very certain about it."	Fallacy of Relevance
The Soviet Union collapsed after taking up atheism. Therefore, we must avoid atheism for the same reasons.	False Causality
You need to go to the party with me, otherwise you'll just be bored at home.	False Dilemma
If we let this child bring the permission slip late, there is no reason to ever set a deadline for anything again!	Faulty Generalization
You can't prove that space aliens don't exist, so they must be real	Intentional

3.2 Data Pre-processing

Each dataset contained text samples with corresponding issue labels. We began by filtering out rows with missing values to ensure consistency and avoid performance issues during modeling. Next, we applied label encoding to convert categorical labels into numerical form, standardizing the format for both datasets.

To prepare the text for model training, we used TF-IDF vectorization with a maximum of 5000 features, capturing the most informative terms while reducing computational load and overfitting. The vectorizer was fitted on the training data and then applied to the test set to maintain a consistent feature space. Finally, we split the training data into 80% for training and 20% for validation.

3.3 Implementation and Evaluation

Baseline. We began by loading a tokenizer and a pre-trained LLM for sequence classification, adjusting the label size to match our dataset classes. The token embeddings were resized to fit the tokenizer's vocabulary, and the end-of-sequence token was set as the padding token to ensure consistent tokenization. Each sample was tokenized to a uniform length of 512 tokens with padding as needed, enabling efficient batching during training. We monitored precision, recall, and F1-score to track performance. Training used a learning rate of 2e-5 over 5 epochs, with weight decay (0.01) and a linear learning rate scheduler for regularization.

Few-shot. We applied a few-shot learning approach by embedding three relevant text-label examples from the training data into each prompt. This format provides contextual guidance, helping the model predict labels for new inputs based on similar cases. These few-shot prompts are tokenized efficiently, ensuring they stay within model limits for smooth batch processing. Each dataset instance includes the tokenized prompt and label tensor, supporting training and evaluation. By integrating context directly into inputs, this method aims to boost model accuracy on unseen data.

Model performance evaluation in different fallacy classes. We generate a confusion matrix that visualizes the performance of a classification model. First, predictions are obtained from the model on the validation dataset. The predicted labels are extracted by selecting the highest probability class from the model's predictions, and the true labels are also collected. The confusion matrix, which compares the true labels with the predicted labels, is then computed. A custom function is then defined to plot the matrix as a heatmap with annotations, where the x-axis represents predicted labels, and the y-axis represents true labels. Finally, the function is called with class names (obtained from the label encoder) and the matrix is saved.

4 Results

Table 2 presents a comparative analysis of baseline (zero-shot) and few-shot prompting across various models on the LOGIC and LOGIC Climate datasets.

Table 2: Comparison of the best results of baseline (zero-shot) and few-shot prompting on LOGIC and LOGIC Climate. The best results per model are **boldfaced** and the overall best results are underlined.

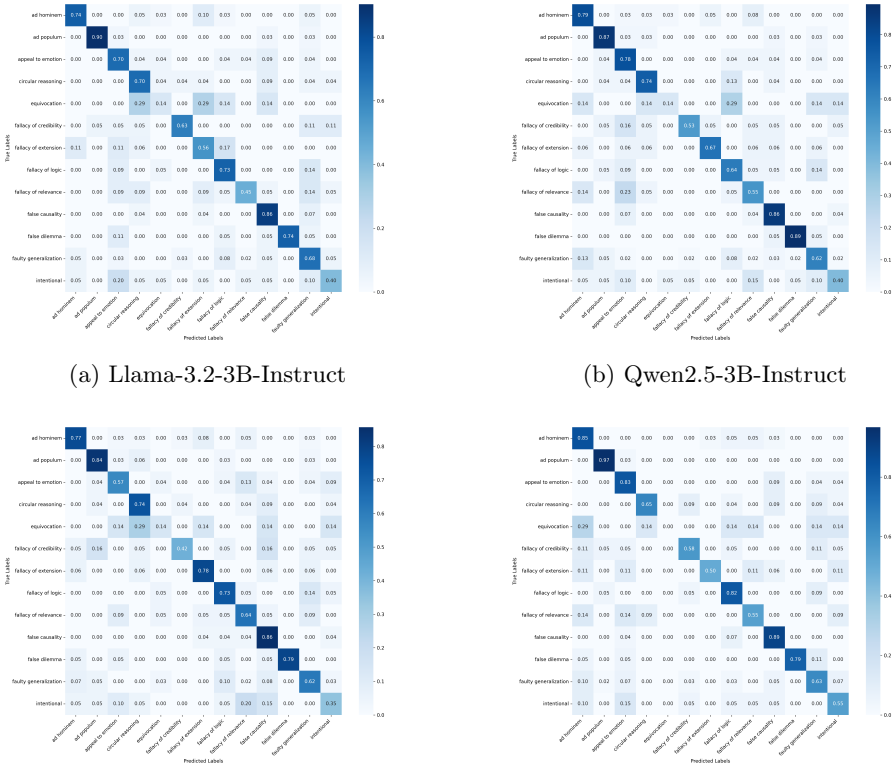
Model	Type	LOGIC				LOGIC Climate			
		A	P	R	F1	A	P	R	F1
Llama-3.2-1B-Instruct	baseline	0.562	0.578	0.562	0.562	0.232	**0.276**	0.232	0.214
	few-shot	**0.631**	**0.646**	**0.631**	**0.632**	**0.261**	0.225	**0.261**	**0.226**
Llama-3.2-3B-Instruct	baseline	0.595	0.616	0.595	0.592	0.213	0.207	0.213	0.191
	few-shot	**0.704**	**0.724**	**0.704**	**0.698**	**0.237**	**0.323**	**0.237**	**0.210**
Qwen2.5-1.5B-Instruct	baseline	0.613	0.631	0.613	0.614	0.251	0.230	0.251	0.232
	few-shot	**0.628**	**0.690**	**0.628**	**0.636**	**0.265**	**0.297**	**0.265**	**0.247**
Qwen2.5-3B-Instruct	baseline	0.601	0.627	0.601	0.600	0.227	**0.263**	0.227	**0.227**
	few-shot	**0.680**	**0.712**	**0.680**	**0.682**	**0.242**	0.260	**0.242**	0.217
Gemma-7b-it	baseline	0.544	**0.648**	0.544	**0.551**	0.171	**0.261**	0.171	**0.137**
	few-shot	**0.550**	0.545	**0.550**	0.543	**0.190**	0.126	**0.190**	0.125
Gemma-2-2b-it	baseline	0.595	0.602	0.595	0.587	0.246	0.267	0.246	0.222
	few-shot	**0.674**	**0.681**	**0.674**	**0.669**	**0.294**	**0.366**	**0.294**	**0.269**
Phi-3-mini-4k-instruct	baseline	0.637	0.665	0.637	0.642	0.237	0.238	0.237	0.216
	few-shot	**0.695**	**0.712**	**0.695**	**0.690**	**0.322**	**0.323**	**0.322**	**0.297**
Phi-3.5-mini-instruct	baseline	0.640	0.682	0.640	0.647	0.251	0.206	0.251	0.222
	few-shot	**0.713**	**0.721**	**0.713**	**0.708**	**0.332**	**0.293**	**0.332**	**0.292**

Accuracy (A), precision (P), recall (R), and F1-score (F1) are among the measures that are assessed. The findings show that few-shot prompting clearly outperforms zero-shot prompting for all models, with notable gains in accuracy and F1-score. The few-shot method shows significant improvements in predicting accuracy and improves the algorithms' capacity to categorize logical fallacies.

Phi-3.5-mini-instruct (few-shot) achieves the highest F1-score (0.708) and accuracy (0.713) on the LOGIC dataset, making it the best-performing model. Llama-3.2-3B-Instruct (few-shot) leads in precision (0.724) and recall (0.704). Other models, like Gemma-2-2b-it (few-shot) and Phi-3-mini-4k-instruct, also perform well with F1-scores above 0.690. Larger models generally outperform baselines and benefit most from few-shot prompting. LOGIC Climate proves more challenging, with lower scores across models. Gemma-2-2b-it (few-shot) ranks highest with 0.332 accuracy and 0.269 F1-score, followed by Phi-3-mini-4k-instruct (few-shot) at 0.322 accuracy and 0.297 F1-score. The results highlight the need for further fine-tuning and dataset-specific adjustments.

There are a number of reasons why certain networks perform poorly on the LOGIC Climate dataset in comparison to the standard LOGIC dataset. There are knowledge gaps since many LLMs are not expressly trained on climate-related talks, which frequently require domain-specific knowledge, scientific terminology, and politically sensitive disputes. It is more challenging to classify logical fallacies in climate discourse since they are typically more nuanced and include cherry-picking evidence, misinterpreting scientific uncertainties, and emotionally laden arguments. The notably poorer accuracy, precision, recall, and F1-scores for all

models point to a basic difficulty in spotting fallacies in arguments pertaining to climate change. Although performance is enhanced by few-shot learning, the benefits are not significant, indicating that adaptive learning is insufficient to make up for domain knowledge deficiencies. While some models, like LLaMA-3.2-1B/3B and Gemma-7B/2B, struggle more, suggesting a lack of adaptation to domain-specific logical reasoning, others, like Phi-3.5-mini-instruct and Phi-3-mini-4k-instruct, do better, probably because they can follow instructions.



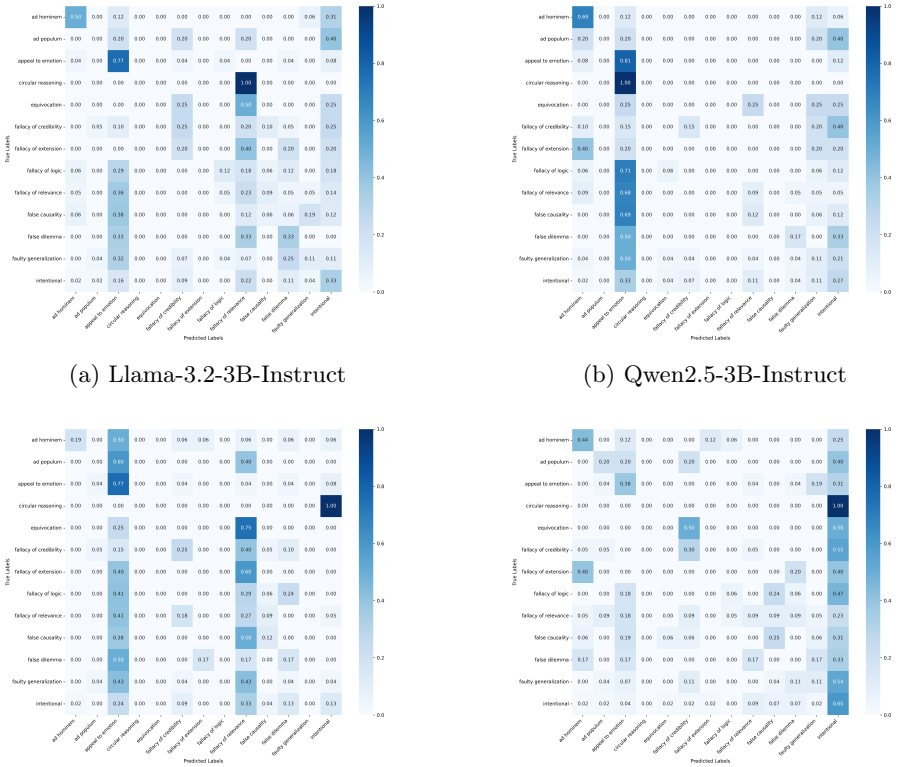
(a) Llama-3.2-3B-Instruct (b) Qwen2.5-3B-Instruct

(c) Gemma-2-2b-it (d) Phi-3.5-mini-instruct

Fig. 1: Performance of LLMS on LOGIC dataset for fallacy classification using a few-shot prompting approach. Each model's performance was visualized with a confusion matrix, providing insights into the classification accuracy across different fallacy types. The normalized confusion matrices for each model illustrate the proportion of correct and misclassified predictions for each fallacy class, allowing for a comparative analysis of model performance in detecting specific logical fallacies.

Figure 1 showed that Llama-3.2-3B-Instruct achieved balanced performance, excelling in "ad populum" (90%) but face difficulties with complex fallacies.

Qwen2.5-3B-Instruct demonstrated consistency in detecting straightforward fallacies, though it struggled with "equivocation" and "fallacy of relevance". Gemma-2-2b-it showed high accuracy in "ad populum" (84%) and "false causality" (86%) but found distinctions between emotional and relevance-based fallacies challenging. Similarly, Phi-3.5-mini-instruct performed well in identifying "ad hominem" (90%), "ad populum" (97%), and "false causality" (89%) but struggled with fallacies like "equivocation" and "intentional". Overall, the models did well on fallacies that are easy to see, such "ad populum" and "false causality," but they struggled with more subtle fallacies that need for interpretive reasoning, like "equivocation" and "intentional".



(a) Llama-3.2-3B-Instruct

(b) Qwen2.5-3B-Instruct

(c) Gemma-2-2b-it

(d) Phi-3.5-mini-instruct

Fig. 2: Performance of LLMs on LOGIC Climate dataset for fallacy classification using few-shot prompting. Each model's performance was visualized through confusion matrices, providing insights into classification accuracy across different fallacy types. The results shed light on each model's strengths and limitations in identifying specific climate-related fallacies.

Figure 2 showed that Llama-3.2-3B-Instruct did well on overt fallacies but struggled to differentiate between more subtle categories like "fallacy of relevance." Similar to this, Qwen2.5-3B-Instruct demonstrated proficiency in "appeal to emotion" and "circular reasoning," but mislabeled sophisticated fallacies such as "fallacy of extension." Gemma-2-2b-it did well on "equivocation" and "circular reasoning" (75%) but had trouble with fallacies that are closely linked, such as "fallacy of relevance". Phi-3.5-mini-instruct performed well in "circular reasoning" (100%) but had trouble with more subtle fallacies like "appeal to emotion" and "ad populum." Overall, all models shown limitations in interpretive reasoning by performing well at identifying "circular reasoning" but poorly at identifying subtle fallacies like "equivocation" and "intentional". Misclassifications that overlap, especially between "ad hominem" and "appeal to emotion," show that linguistic similarities in climate-related fallacies pose a problem to LLMs.

5 Discussion

5.1 Dataset

The majority of the examples in the datasets we used had one or two sentences, with the rare paragraph. However, real life usually consists of a few phrases or a few paragraphs. Although our analysis demonstrates that the LLMs functions for shorter texts. Lengthier texts—which are likely to contain a greater number of logical fallacies—may not yield the same results. To the best of our knowledge, there is no labeled dataset on logical fallacies for lengthy texts, thus generating one by hand would be necessary. Nevertheless, this would still warrant additional testing.

5.2 Prompt engineering

Even though we designed a straightforward prompt to enhance the model's few-shot learning capabilities and provide context when processing inputs, different prompt constructions may impact the outcome.

5.3 Misclassifications

The confusion matrices shed more light on the shortcomings of the models, especially when it comes to differentiating semantically related fallacies like appeal to emotion, circular reasoning, and fallacy of relevance. All models, regardless of size, exhibited confusion between these categories, highlighting the difficulty of detecting fallacies that share overlapping linguistic or argumentative features. This implies that existing models struggle with smaller distinctions, which probably led to the poorer recall scores seen in numerous models, even while they can handle fallacies with more distinguishing properties (like ad hominem and ad populum).

6 Limitations

First off, the datasets we used are mostly composed of brief text samples, usually consisting of just one or two phrases. Fallacies are frequently included into longer discourses, sometimes spanning several paragraphs, in real-world settings. As a result, it is unclear how well the models will perform on longer, more complicated texts. The thorough evaluation of model performance in such scenarios would require the creation of a labeled dataset with additional arguments.

Second, the few-shot prompting strategy depends largely on prompt construction, even though it can be successful in some situations. The prompt's design has a big impact on the model's output, and different prompt formulations may provide different outcomes. Even though we followed a set pattern, trying out different prompt designs could result in even better results. Optimized prompt engineering methods should be investigated in future studies to improve model accuracy, especially for subtle fallacies.

Lastly, the models had trouble differentiating between fallacies that share semantic similarities, including "appeal to emotion," "circular reasoning," and "fallacy of relevance." A shortcoming in the models' interpretive reasoning was highlighted by the numerous misclassifications caused by the overlapping linguistic and argumentation elements in these categories. These difficulties imply that in order to enhance their capacity to correctly distinguish closely similar logical fallacies, current LLMs could need more instruction or practice with a wider range of fallacy cases. Addressing these limitations can lead to advancing LLM capabilities for reliable logical fallacy detection in real-world applications.

7 Conclusions and Future Work

This study provides a comprehensive evaluation of several open source LLMs for detecting logical fallacies, using the LOGIC and LOGIC Climate datasets as benchmarks. We examined how well they performed in recognizing particular fallacy kinds using confusion matrices and common assessment metrics. According to our research, these models are effective at identifying simpler fallacies like "circular reasoning," but they have trouble identifying more complex ones like "equivocation" and "intentional," which call for more complex interpretive reasoning. The inability of current LLMs to handle subtle differences in logical arguments was highlighted by the frequent misclassifications caused by overlapping features among several fallacies. These findings imply that although LLMs may be useful for fallacy detection tasks, more work is necessary to ensure consistent accuracy, particularly in intricate, real-world texts.

Future research should explore advanced prompt engineering and fine-tuning techniques to enhance LLMs' interpretive capabilities. Furthermore, creating more extensive, contextually rich datasets may help give the models a stronger basis on which to detect errors in lengthy discourse. By resolving these issues, LLMs may develop into useful instruments for critical thinking exercises, enhancing the integrity and quality of information in a variety of fields, including media literacy, education, and automated content moderation.

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